

Closed Random Sublevels Force Epigraph Closure in Two Broad Regimes

A partial answer to Remark 2.14 of *On random convex analysis*

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Status. Candidate partial result, likely valid, pending human review. The full arbitrary-net question remains open here.

1 Source question

Guo, Zhang, Wu, Yang, Yuan, and Zeng ask the following in Remark 2.14 of [1], PDF page 10. Let $(E, \mathcal{P})$ be a random locally convex (RLC) module over a sigma-finite measure space $(\Omega, \mathcal{F}, \mu)$, and let $f : E \rightarrow \bar{L}^0(\mathcal{F})$ be proper and local. Compare

- (1)' $\{x \in E : f(x) \leq r\}$ is $\mathcal{T}_{\varepsilon, \lambda}$ -closed for every $r \in L^0(\mathcal{F})$,
- (2)' $\text{epi}(f) = \{(x, r) : f(x) \leq r\}$ is closed in $(E, \mathcal{T}_{\varepsilon, \lambda}) \times (L^0(\mathcal{F}), \mathcal{T}_{\varepsilon, \lambda})$.

The authors observe $(2)' \Rightarrow (1)'$ and state that they do not know whether $(1)' \Rightarrow (2)'$ in general. Their Theorem 2.13 gives the implication when the module and seminorm family have the countable concatenation property.

2 Idea of the proof

The obstruction posed by random thresholds disappears for sequences. If $r_n \rightarrow r$ in probability, a subsequence converges almost everywhere. Its pointwise supremum is consequently a finite random variable. On any positive-measure set where $f(x)$ lies uniformly above r , one can discard a finite initial segment so that this supremum remains strictly below $f(x)$ on a positive subset. All corresponding x_n then lie in one fixed random sublevel, whose sequential closedness gives the contradiction.

This mechanism yields sequential epigraph closure on every RLC module. When the module topology is first countable, sequential closure is ordinary closure, proving the requested implication without countable concatenation.

There is a second mechanism for arbitrary closure nets. If the module E has the countable concatenation property, closedness of one fixed sublevel provides a separating neighborhood involving only finitely many seminorms. One selects countably many epigraph points with summable errors for those seminorms, partitions the base by the sets on which their thresholds lie below the fixed level, and pastes the points along that partition. This proves the implication without any countable-concatenation hypothesis on the seminorm family $\mathcal{P}$.

implies that $\lim_{\alpha} f(x_{\alpha}) \leq r$.

Up to now, we have proved that $\lim_{\alpha} f(x_{\alpha}) \geq r$ in either case in which $\mathcal{A}_r$ is empty or nonempty. Since r is an arbitrarily chosen element such that $f(x) > r$ on Ω , we have $\lim_{\alpha} f(x_{\alpha}) \geq f(x)$.

This completes the proof $\square$

Remark 2.14. Let $(E, \mathcal{P})$ be an RLC module over R with base $(\Omega, \mathcal{F}, \mu)$ and $f : E \rightarrow \bar{L}^0(\mathcal{F})$ a proper and local function. Let us observe the following three statements:

- (1)' $\{x \in E \mid f(x) \leq r\}$ is $\mathcal{T}_{\varepsilon, \lambda}$ -closed for all $r \in L^0(\mathcal{F})$;
- (2)' $\text{epi}(f)$ is closed in $(E, \mathcal{T}_{\varepsilon, \lambda}) \times (L^0(\mathcal{F}), \mathcal{T}_{\varepsilon, \lambda})$;
- (3)' $\lim_{\alpha} f(x_{\alpha}) \geq f(x)$ for all $x \in E$ and all net $\{x_{\alpha}, \alpha \in \Gamma\}$ convergent to x with respect to $\mathcal{T}_{\varepsilon, \lambda}$.

Generally, one always has $(3)' \Rightarrow (2)' \Rightarrow (1)'$. Although under the assumption of Theorem 2.13 one also has $(1)' \Leftrightarrow (2)'$, $(1)'$ or $(2)'$ did not imply $(3)'$, in fact, we can construct examples showing that $(3)'$ is not necessarily true even for a $\mathcal{T}_{\varepsilon, \lambda}$ -continuous function. Generally, we do not know if $(1)'$ implies $(2)'$, either, but $(2)'$ as the definition of a $\mathcal{T}_{\varepsilon, \lambda}$ -lower semicontinuous function has met the needs of the study of $\mathcal{T}_{\varepsilon, \lambda}$ -lower semicontinuous functions, see, for instance, [GY12, GZZ15a]. Whereas, for the case of the locally L^0 -convex topology $\mathcal{T}_c$, since we are often forced to assume that RLC module $(E, \mathcal{P})$ in question satisfy the hypothesis of Theorem 2.13 we adopt the general definition of a $\mathcal{T}_c$ -lower semicontinuous function as Definition 2.12. But, in [GZZ15a] we used (3) of Theorem 2.13 as the notion of a $\mathcal{T}_c$ -lower semicontinuous function since we then did not know if (1) of Theorem 2.13 indeed implies (4), and hence also (3), of Theorem 2.13.

Now, owing to Theorem 2.13 we can make perfect the Fenchel–Moreau duality theorem for a $\mathcal{T}$ -lower semicontinuous function. From now on we always use

Figure 1: Remark 2.14 in the source paper (PDF page 10).

3 The sequential theorem

We use the standard fact that the $\mathcal{T}_{\varepsilon, \lambda}$ topology on $L^0(\mathcal{F})$ is metrizable and is convergence in probability for an equivalent probability measure $\hat{\mu}$ (equivalently, convergence locally in measure for the original sigma-finite measure), as recalled in [1, Section 2].

Lemma 1 (random upper envelope). *Let $r_n, r \in L^0(\mathcal{F})$ and $r_n \rightarrow r$ in $\mathcal{T}_{\varepsilon, \lambda}$. Let $B \in \mathcal{F}$ have positive measure and let $\delta > 0$. There are a subsequence (r_{n_j}) , an integer N , a positive-measure set $D \subset B$, and $q \in L^0(\mathcal{F})$ such that*

$$r_{n_j} \leq q \quad (j \geq N), \quad q \leq r + \delta \quad \text{on } D.$$

Proof. Pass to a subsequence, still denoted (r_{n_j}) , which converges to r $\hat{\mu}$ -almost everywhere. For $N \geq 1$, put

$$D_N = B \cap \bigcap_{j \geq N} \{r_{n_j} \leq r + \delta\}.$$

Modulo a null set, $D_N \uparrow B$, so D_N has positive measure for some N . After fixing such an N , set

$$q = \sup_{j \geq N} r_{n_j}.$$

The supremum is measurable and finite almost everywhere because the underlying sequence converges almost everywhere. Thus $q \in L^0(\mathcal{F})$; it dominates every retained r_{n_j} , and $q \leq r + \delta$ on D_N . $\square$

Theorem 2 (closed sublevels imply sequentially closed epigraph). *Let X be any topological space and $f : X \rightarrow \overline{L}^0(\mathcal{F})$. Suppose that for every $q \in L^0(\mathcal{F})$ the sublevel*

$$C_q = \{x \in X : f(x) \leq q\}$$

is sequentially closed in X . Then $\text{epi}(f)$ is sequentially closed in $X \times (L^0(\mathcal{F}), \mathcal{T}_{\varepsilon, \lambda})$.

Proof. Let $(x_n, r_n) \in \text{epi}(f)$ and suppose $(x_n, r_n) \rightarrow (x, r)$. Thus $f(x_n) \leq r_n$ for all n . Assume for a contradiction that $f(x) \not\leq r$. Then $\{f(x) > r\}$ has positive measure. By taking a countable union of uniform-gap sets, choose $\delta > 0$ and a positive-measure set B such that

$$f(x) > r + 2\delta \quad \text{on } B.$$

Apply Lemma 1. For the resulting subsequence, tail, set D , and random variable q , we have

$$f(x_{n_j}) \leq r_{n_j} \leq q \quad (j \geq N).$$

Hence $x_{n_j} \in C_q$ for all $j \geq N$. This subsequence still converges to x , and C_q is sequentially closed, so $f(x) \leq q$. On D , however,

$$q \leq r + \delta < f(x),$$

a contradiction on a set of positive measure. Therefore $f(x) \leq r$ and $(x, r) \in \text{epi}(f)$. $\square$

Theorem 2 is stronger than needed in two respects: it requires neither an L^0 -module structure nor properness or locality of f .

4 The countable-concatenation theorem

Theorem 3 (the module CCP alone is sufficient). *Let $(E, \mathcal{P})$ be an RLC module and suppose that E has the countable concatenation property; no such assumption is made on $\mathcal{P}$. Let $f : E \rightarrow \overline{L}^0(\mathcal{F})$ be proper and local. If every sublevel $C_q = \{f \leq q\}$, $q \in L^0(\mathcal{F})$, is $\mathcal{T}_{\varepsilon, \lambda}$ -closed, then $\text{epi}(f)$ is closed in $(E, \mathcal{T}_{\varepsilon, \lambda}) \times (L^0(\mathcal{F}), \mathcal{T}_{\varepsilon, \lambda})$.*

Proof. Write $\hat{\mu}$ for the probability measure defining $\mathcal{T}_{\varepsilon, \lambda}$. Suppose that (x, r) belongs to the closure of $\text{epi}(f)$ but not to the epigraph. There are a set $B \in \mathcal{F}$ of positive measure and a deterministic $\delta > 0$ such that

$$f(x) > r + 2\delta \quad \text{on } B.$$

Choose $u \in \text{dom}(f)$ and localize outside B by putting

$$\bar{x} = I_B x + I_{B^c} u, \quad \bar{r} = I_B r + I_{B^c} f(u), \quad q = I_B(r + \delta) + I_{B^c} f(u).$$

Locality shows that localization sends epigraph points to epigraph points, and multiplication by indicators is continuous. Hence $(\bar{x}, \bar{r})$ is still in the closure of $\text{epi}(f)$. Moreover $f(\bar{x}) > q$ on B , so $\bar{x} \notin C_q$.

Since C_q is closed, choose a finite nonempty $Q \subset \mathcal{P}$ and $\varepsilon, \lambda > 0$ such that

$$\bar{x} + N_0(Q, \varepsilon, \lambda) \quad \text{is disjoint from } C_q, \tag{1}$$

where

$$N_0(Q, \varepsilon, \lambda) = \{v : \hat{\mu}(\{\|v\|_Q < \varepsilon\}) > 1 - \lambda\}.$$

Choose positive numbers η_n with $\eta_n \downarrow 0$ and $\sum_n \eta_n < \lambda$. Using the closure of the epigraph and then applying the preceding localization, choose $(z_n, t_n) \in \text{epi}(f)$ such that

$$\widehat{\mu}\{\|z_n - \bar{x}\|_Q \geq \varepsilon\} < \eta_n, \quad (2)$$

$$\widehat{\mu}\{|t_n - \bar{r}| \geq \delta\} < \eta_n, \quad (3)$$

and such that $z_n = u$ and $t_n = f(u)$ on B^c . Let $G_n = \{t_n \leq q\}$. The equality on B^c and (3) give

$$\widehat{\mu}(G_n^c) < \eta_n.$$

Consequently $\bigcup_n G_n = \Omega$ modulo null sets, since the measure of $\bigcap_n G_n^c$ is at most $\inf_n \eta_n = 0$. Define the measurable partition

$$D_1 = G_1, \quad D_n = G_n \setminus \bigcup_{k < n} G_k \quad (n \geq 2).$$

The CCP of E supplies

$$y = \sum_{n=1}^{\infty} I_{D_n} z_n \in E.$$

For each n , locality gives $I_{D_n} f(y) = I_{D_n} f(z_n)$. Because $D_n \subset G_n$ and $f(z_n) \leq t_n$, it follows that $f(y) \leq q$; hence $y \in C_q$. On the other hand, locality of every random seminorm gives

$$\{\|y - \bar{x}\|_Q \geq \varepsilon\} = \bigcup_n (D_n \cap \{\|z_n - \bar{x}\|_Q \geq \varepsilon\}) \quad \text{modulo null sets.}$$

By (2), its probability is at most $\sum_n \eta_n < \lambda$. Thus $y \in \bar{x} + N_0(Q, \varepsilon, \lambda)$, contradicting (1). $\square$

Source Theorem 2.13 assumes that both E and $\mathcal{P}$ have the countable concatenation property. Theorem 3 removes the assumption on $\mathcal{P}$ for the implication $(1)' \Rightarrow (2)'$.

5 Affirmative subcases of Remark 2.14

Corollary 4 (first-countable case). *Let $(E, \mathcal{P})$ be an RLC module such that $(E, \mathcal{T}_{\varepsilon, \lambda})$ is first countable. For any map $f : E \rightarrow \overline{L}^0(\mathcal{F})$, if every random sublevel $\{f \leq q\}$ is $\mathcal{T}_{\varepsilon, \lambda}$ -closed, then $\text{epi}(f)$ is closed in $(E, \mathcal{T}_{\varepsilon, \lambda}) \times (L^0(\mathcal{F}), \mathcal{T}_{\varepsilon, \lambda})$.*

Proof. Closed sublevels are sequentially closed, so Theorem 2 makes the epigraph sequentially closed. The factor $L^0(\mathcal{F})$ is metrizable, and the product of a first-countable space with a metrizable space is first countable. In a first-countable space, sequentially closed sets are closed. $\square$

Corollary 5 (countably generated RLC modules). *The implication $(1)' \Rightarrow (2)'$ in Remark 2.14 holds, without any countable concatenation assumption, if the $\mathcal{T}_{\varepsilon, \lambda}$ topology of E is generated by a countable family of random seminorms. In particular it holds for every random normed module.*

Proof. For a countable seminorm family, finite subfamilies together with positive rational values of ε and λ give a countable local base for $\mathcal{T}_{\varepsilon, \lambda}$. Thus Corollary 4 applies. A random normed module has a singleton defining seminorm family. $\square$

6 Why the full net problem remains

When E lacks the CCP, the proof of Theorem 3 produces only a formal countable paste, which need not be an element of E . The sequential proof also cannot simply replace “subsequence” by “subnet.” A net may converge in probability while carrying arbitrarily high values on measurable sets whose probabilities shrink and whose locations move with the index. A whole tail then need not possess a finite pointwise random upper bound. Moreover, for an uncountable seminorm family the $\mathcal{T}_{\varepsilon,\lambda}$ topology of E need not be first countable or sequential, so a point in the closure of the epigraph need not be witnessed by an E -convergent sequence.

Thus the theorems settle the sequential content of the source question, all first-countable cases, and all modules E with the CCP (regardless of $\mathcal{P}$). They neither prove nor disprove the implication in the remaining regime where E lacks the CCP and its topology is genuinely nonsequential. This is why the packet remains classified as a partial result.

7 Verification and novelty status

The proofs use only: (i) the elementary almost-everywhere subsequence principle for convergence in probability; (ii) finiteness of the pointwise supremum of an almost-everywhere convergent real-valued sequence; and (iii) the first-countable characterization of closed sets by sequences; and (iv) locality plus summable probability errors for the countable-paste argument. No computational or unproved dependency is present.

The run’s lightweight indexes and local arXiv corpus were searched for the paper id, the exact Remark 2.14 question, and close epigraph/sublevel and countable-concatenation terminology. Bounded web searches on 9 August 2026 found the source and background papers but no later resolution of either the stated subcase or the full question. This is evidence of plausible novelty, not a comprehensive certification.

Human-review focus. Verify Lemma 1, especially the positive-measure tail set and the almost-everywhere finiteness of q ; verify the localization and good-set partition in Theorem 3; check the summable seminorm-error estimate for its pasted point; and verify that the chosen seminorm hypothesis indeed supplies a countable $\mathcal{T}_{\varepsilon,\lambda}$ local base.

References

- [1] T. Guo, E. Zhang, M. Wu, B. Yang, G. Yuan, and X. Zeng, *On random convex analysis*, arXiv:1603.07074 (2016), especially Section 2 and Remark 2.14.